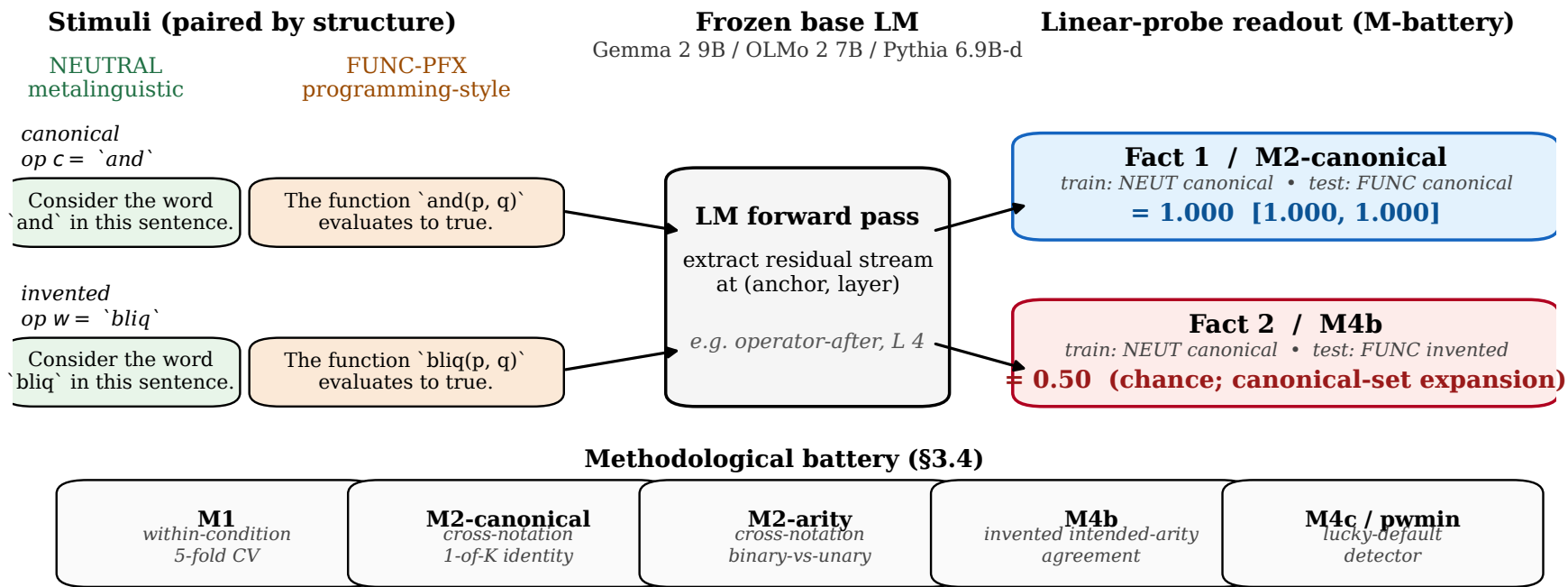




All three model families return Fact 1 = ceiling and Fact 2 = chance at the principal operator-after L 4 cell under the v6 scope (15 canonicals × 16 invented). The §4.1.1 content-word control extends Fact 1 to in-vocabulary content words (trained-vocabulary substrate-invariance, not specifically operator-class).

Figure 1. Schematic of the substrate-invariance setup. Notations / probes / measurements at a glance.